

EXPERIMENT 3

Design of Test Cases and Maintenance Plan for a College Management System

Aim

To design comprehensive test cases for the major modules of a College Management System and prepare a maintenance plan to ensure software quality, security, and reliability throughout its lifecycle.

Objective

- Design functional test cases.
- Verify expected system behavior.
- Plan software maintenance activities.
- Improve software quality and user satisfaction.

TC ID	Module	Test Scenario	Expected Result	Status
TC01	Login	Valid credentials	Dashboard opens	Pass
TC02	Registration	Valid student details	Registration successful	Pass
TC03	Attendance	Faculty marks attendance	Attendance saved	Pass
TC04	Examination	View result	Correct result displayed	Pass
TC05	Fee	Online fee payment	Receipt generated	Pass
TC06	Library	Issue a book	Book issued	Pass

Introduction

Software testing validates whether an application satisfies its functional and non-functional requirements. A well-designed testing strategy identifies defects before deployment and ensures that every software module performs correctly. After deployment, maintenance activities keep the application updated, secure, and compatible with changing business requirements.

Sample Application

The College Management System is a web-based application that automates admissions, attendance, examination management, fee collection, library services, student records, and faculty management. The system provides separate interfaces for students, faculty members, librarians, and administrators.

Testing Objectives

Verify functionality, improve reliability, validate security, detect defects early, reduce maintenance cost, and ensure user satisfaction.

Types of Testing

Unit Testing: Tests individual modules.

Integration Testing: Ensures modules communicate correctly.

System Testing: Validates the complete system.

User Acceptance Testing: Confirms user requirements.

Security Testing: Verifies authentication and authorization. Performance

Testing: Measures response time under load.

Test Case Design

TC07	Faculty	Upload marks	Marks saved	Pass
TC08	Login	Invalid password	Error displayed	Pass

Maintenance Plan

Corrective Maintenance: Fix defects reported after deployment.

Adaptive Maintenance: Modify software to meet new academic regulations or operating environments.

Perfective Maintenance: Improve performance, usability, and efficiency based on user feedback.

Preventive Maintenance: Refactor code, optimize the database, and strengthen security to prevent future failures.

Maintenance Schedule

Activity	Frequency
Bug Fixes	As Required
Database Backup	Daily
Performance Monitoring	Weekly
Security Updates	Monthly
Software Upgrade	Every Semester
User Feedback Analysis	Monthly

Tools Used

Selenium, JUnit/PyTest, MySQL, GitHub, Jira, Apache JMeter.

Benefits of Testing

Testing reduces software defects, increases reliability, improves security, minimizes maintenance costs, and enhances customer satisfaction.

Expected Outcome

The designed test cases ensure that every module of the College Management System functions correctly. The maintenance plan guarantees long-term software reliability, security, performance, and adaptability.

Conclusion

Software testing and maintenance are essential for delivering high-quality applications. The designed test cases validate critical modules of the College Management System, while the maintenance plan supports continuous improvement through bug fixing, security enhancements, and feature updates. Together, these activities increase software quality, reduce operational risks, and improve the overall user experience.